

CAMERON GILREATH

Orlando/Tallahassee, FL
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EDUCATION

Florida Agricultural and Mechanical University | Tallahassee, FL | Expected May 2029

Bachelor of Science in Electrical Engineering; 3.31/4.0 GPA

Relevant Coursework: Calculus I & II, Programming Fundamentals, First Year Engineering Lab, Chemistry I, Chemistry Lab, Trigonometry, Environmental Science, Communications I & II

TECHNICAL AND PROFESSIONAL SKILLS

Technical: MATLAB/SIMULINK, Python, C++, Arduino (Embedded C), JavaScript, Web Development

Professional: Technical Proficiency, Leadership, Adaptability, Communication, Problem Solving

ACADEMIC EXPERIENCE & PROJECTS

Personal Project - Arduino Security System | June 2026

- Developed a WiFi Security Alarm using an ESP8266 microcontroller and ultrasonic sensor, implementing automated network reconnection logic, local hardware alerts, and HTTP-based data dispatching.
- Built a Multi-Threaded Desktop Interface and Backend using Flask and Tkinter, integrating an SMTP email notification system with a custom cooldown algorithm to optimize alert frequency.

Electrical Modeling of My Vehicle's Audio System | March 2026

- Recreated my vehicle's car-audio power and signal architecture in MATLAB/Simulink, modeling real-world DC distribution, grounding, fuse protection, RCA routing, and multi-channel amplification
- Built detailed amplifier, subwoofer, and full-range speaker models using circuit analysis, control techniques, and DSP principles to evaluate realistic loading, filtering, and distortion behavior

Insulin Pump Controller using PID Control | February 2026

- Collaborated in a team of 4 as the Electrical Engineer on a multi-disciplinary team to design and prototype a closed-loop insulin pump controller using an Arduino Uno to automate glucose delivery
- Optimized a PID control system through rigorous testing of proportional, integral, and derivative gains to achieve a target glucose setpoint of 70 mg/dL with 0% overshoot, ensuring patient safety

Personal Project - Solar Powered Portable Speaker | July 2024

- Engineered and assembled a high-output solar-powered Bluetooth speaker by integrating a 12V battery, solar panel, and charging station into a custom-modified cooler housing.
- Integrated and connected electronic components, including a Bluetooth amplifier, subwoofer, and car speakers, using a 3-prong switch for efficient power management and signal distribution.

WORK & LEADERSHIP EXPERIENCE

Cam's Cellular Care, LLC | Orlando, FL | July 2020 - Present

Chief Executive Officer

- Provided service repairs to over 150 Apple devices
- Developed full-stack web solutions to enhance digital presence for commercial clients
- Oversaw all accounting operations, including accounts payable, accounts receivable, and payroll

Universal Orlando | Orlando, FL | May 2025 - August 2025

Quick Service Associate

- Provided excellent customer service to hundreds of international customers daily
- Prepared and served hundreds of guests with food and beverages
- Orchestrated ten different carts with different food options which called for quick adjustments

ACTIVITIES & HONORS

Progressive Black Men, Inc | Symphonic Band | Institute of Electrical and Electronic Engineers | National Society of Black Engineers (NSBE) | Orlando Club at FAMU | Alpha Kappa Alpha Scholarship Pageant | Entrepreneurship Club | Dean's List 2025 | Honor Roll 2026